

# Scale-Dependent Information Units as Phase Transitions on $T^4$

Geometry as the Primary Foundation of Information and  
Energy

Johann Pascher

May 2026

Dok. 255 derived Landauer's principle and Vopson's mass-information equivalence from  $T^4$  geometry, identifying information with topologically frozen kinetic energy. The present refinement shows that this derivation holds as a special case at the thermodynamic scale, not universally. The more fundamental picture is: the minimal information unit is by definition the topological winding quantum  $\Delta w = 1$  — scale-universal and energy-independent. The corresponding energy  $E_{\text{bit}}(L) = \hbar c/L$  follows from action and the uncertainty relation and is scale-specific. Landauer's  $E_L = k_B T \ln 2$  is the special case at the temperature-length scale where  $E_{\text{bit}}(L) = k_B T \ln 2$  — biological room temperature. All phase transitions are discrete because  $\pi_1(T^4) \cong \mathbb{Z}^4$  admits no half-integer winding quanta. Geometry is more fundamental than energy. We work in natural units with  $c = \hbar = k_B = 1$  unless stated otherwise.

## Contents

### 1 The Problem: Three Levels, One Hierarchy

3

|          |                                                          |          |
|----------|----------------------------------------------------------|----------|
| <b>2</b> | <b>Three Independent Levels</b>                          | <b>3</b> |
| 2.1      | Visualisation of the Hierarchy . . . . .                 | 4        |
| <b>3</b> | <b>Geometry is More Fundamental than Energy</b>          | <b>5</b> |
| <b>4</b> | <b>Phase Transitions are Discrete at Every Scale</b>     | <b>5</b> |
| <b>5</b> | <b>Scale Resonance at the Scale of Life — Hypothesis</b> | <b>5</b> |
| <b>6</b> | <b>Refinement of Dok. 255</b>                            | <b>6</b> |
| <b>7</b> | <b>Summary</b>                                           | <b>6</b> |

# 1 The Problem: Three Levels, One Hierarchy

Dok. 255 stated:

*Information is topologically frozen kinetic energy.*

This is correct as a description of the thermodynamic regime. The deeper question is: what is more fundamental — geometry or energy?

The internal consistency check shows that the equivalence temperature at which  $E_{\text{bit}}(L) = k_B T \ln 2$  holds varies over **70 orders of magnitude**:

| Scale       | L [m]                                  | $T_{\text{equiv}}$ [K] |
|-------------|----------------------------------------|------------------------|
| Planck      | $1.6 \times 10^{-35}$                  | $2.0 \times 10^{32}$   |
| FFGFT $L_0$ | $2.1 \times 10^{-39}$                  | $1.5 \times 10^{36}$   |
| QCD/Proton  | $1.0 \times 10^{-15}$                  | $3.3 \times 10^{12}$   |
| Atom        | $1.0 \times 10^{-10}$                  | $3.3 \times 10^7$      |
| DNA codon   | $1.0 \times 10^{-9}$                   | $3.3 \times 10^6$      |
| <b>Cell</b> | <b><math>1.0 \times 10^{-5}</math></b> | <b>331</b>             |
| Cosmic      | $4.4 \times 10^{26}$                   | $7.5 \times 10^{-30}$  |

with the formula (cf. Section 2):

$$T_{\text{equiv}}(L) = \frac{\hbar c}{k_B L \ln 2}. \quad (1)$$

Only at the cellular scale does  $T_{\text{equiv}} \approx 331$  K match biological room temperature.

## 2 Three Independent Levels

The calculation enforces a separation into three levels:

### Level 1: Topological (universal).

**Definition 1** (Minimal information unit). *The minimal information unit on  $T^4$  is the winding quantum  $I_{\text{bit}} = \Delta w = 1$ ,  $\Delta w \in \mathbb{Z}$ . One bit is identified with a minimal topological winding difference  $\Delta w = 1$ .*

This identification is the conceptual bridge between topology and information theory. The unit is scale-independent:  $\pi_1(T^4) \cong \mathbb{Z}^4$  enforces discrete integer winding numbers at every scale. No phase transition can produce half a winding quantum.

**Level 2: Geometric (scale-specific).** The energy of a winding quantum at length scale  $L$  follows from action and the uncertainty relation: The action of one winding quantum is  $S = \Delta w \cdot h = h$ . From the uncertainty relation  $\Delta x \cdot \Delta p \sim \hbar$  at spatial scale  $L$  one obtains  $p \sim \hbar/L$ , hence:

$$E_{\text{bit}}(L) = pc = \frac{\hbar c}{L}. \quad (2)$$

*Note:* The numerical prefactor is a scale definition; the formula gives the correct order of magnitude. In natural units ( $c = \hbar = k_B = 1$ ),  $E_{\text{bit}}(L) = 1/L$ .

**Level 3: Thermodynamic (one scale).** Landauer's principle [2]:

$$E_L = k_B T \ln 2 \quad \text{per bit} \quad (3)$$

holds in the thermodynamic regime, i.e. when  $E_{\text{bit}}(L) = E_L$ . Setting equal yields equation (1).

## 2.1 Visualisation of the Hierarchy

| Level | Character     | Quantity                        | Validity                           |
|-------|---------------|---------------------------------|------------------------------------|
| 1     | topological   | $\Delta w = 1$                  | universal, all scales              |
|       | ↓             |                                 |                                    |
| 2     | geometric     | $E_{\text{bit}}(L) = \hbar c/L$ | every scale $L$                    |
|       | ↓             |                                 |                                    |
| 3     | thermodynamic | $E_L = k_B T \ln 2$             | only when $E_{\text{bit}} = k_B T$ |

### 3 Geometry is More Fundamental than Energy

The derivation hierarchy is:

$$\begin{aligned} T^4\text{-geometry} &\longrightarrow \Delta w = 1 \longrightarrow E_{\text{bit}}(L) = \hbar c/L \\ &\longrightarrow E_L = k_B T \ln 2 \text{ (special case)}. \end{aligned} \quad (4)$$

Energy is a scale-dependent projection of  $T^4$  geometry, not its foundation.

---

**Dok. 255** Information = frozen kinetic energy

**Dok. 257** Information = winding quantum  $\Delta w$ ; energy  
is a scale-specific projection of it

---

*Both statements consistent: Dok. 255 describes the thermodynamic special case.*

---

### 4 Phase Transitions are Discrete at Every Scale

All phase transitions in physics are discrete. This follows directly from  $\pi_1(T^4) \cong \mathbb{Z}^4$ .

| Scale        | Phase transition           | Information unit                            |
|--------------|----------------------------|---------------------------------------------|
| Planck/FFGFT | Topological enclosure      | $\Delta w = 1, E \sim E_P$                  |
| QCD          | Confinement, hadronisation | Baryon $B = 1, E \sim \Lambda_{\text{QCD}}$ |
| Atom         | Electron configuration     | Quantum number $n, E \sim \text{eV}$        |
| Molecule/DNA | Codon, folding             | 3 base pairs, $E \sim k_B T$                |
| Cell         | Protein folding            | Domain, $E \sim k_B T$                      |
| Cosmic       | $\sigma$ -shift            | $\hbar c/R_U, E \ll k_B T_{\text{CMB}}$     |

At every scale:  $I_{\text{bit}} = \Delta w = 1$  (topological). The energy of the bit varies with scale.

### 5 Scale Resonance at the Scale of Life — Hypothesis

For  $L \sim 10^{-5}$  m (cell size) and  $T \sim 300$  K:

$$E_{\text{bit}}(L_{\text{cell}}) = \frac{\hbar c}{10^{-5} \text{ m}} \approx 3.2 \times 10^{-21} \text{ J} \approx k_B \cdot 331 \text{ K} \cdot \ln 2. \quad (5)$$

This suggests the following hypothesis:

*Life as a self-organising, information-processing structure  
may only exist at scales where  $E_{\text{bit}}(L) \approx k_B T$ .  
Biological room temperature would then be a consequence of  
this scale resonance.*

Whether this coincidence is causal or accidental remains open and deserves further investigation. It is stated here as a hypothesis, not a derivation.

## 6 Refinement of Dok. 255

1. **Correct in Dok. 255:** Landauer and Vopson follow from  $T^4$  geometry in the thermodynamic regime.
2. **Refinement:** The universal information unit is  $\Delta w = 1$  (topological), not  $k_B T \ln 2$  (thermodynamic).
3. **Extension:** Energy is a scale-specific projection of geometry.
4. **Consequence:**  $I_{\text{bit}} = \Delta w$  holds at all scales.  $E_L = k_B T \ln 2$  holds only at the scale of life.  $E_{\text{bit}} = \hbar c/L$  holds geometrically at every scale.

## 7 Summary

1. The minimal information unit is by definition the winding quantum  $\Delta w = 1$  — universal from  $\pi_1(T^4) \cong \mathbb{Z}^4$ .
2. The energy of a bit follows from  $S = h$  and  $\Delta x \Delta p \sim \hbar$ :  $E_{\text{bit}}(L) = \hbar c/L$ .
3. Landauer  $E_L = k_B T \ln 2$  is the special case at biological room temperature ( $L \sim 10^{-5} \text{ m}$ ,  $T \sim 300 \text{ K}$ ).
4. All phase transitions are discrete because  $\pi_1(T^4) \cong \mathbb{Z}^4$  is integer-valued.
5. Geometry is more fundamental than energy:  $T^4 \rightarrow \Delta w \rightarrow E_{\text{bit}}(L) \rightarrow E_L$  (special case).

6. Scale resonance at the scale of life is an open hypothesis, not a derivation.

**Methodological classification:** Dok. 257 is an algebraic bridge from  $T^4$  and  $\pi_1(T^4) \cong \mathbb{Z}^4$ . The scale resonance hypothesis requires further investigation.

**Integration in the corpus.** The three-level hierarchy formulated here is the field-theoretically developed form of the static/dynamic demarcation of Doc. 261: level 1 (topological,  $\Delta w$ ) is layer 1 – the static, reference-free  $\xi$ -geometry level. Levels 2 and 3 ( $E_{\text{bit}}(L) = \hbar c/L$ ;  $E_L = k_B T \ln 2$ ) are layer-2 projections and therefore carry the embedding constants  $\hbar, c, k_B$ . The geometric formula  $E_{\text{bit}}(L) = \hbar c/L$  coincides with the general energy hierarchy  $E_N = \hbar c/L_N$  of Doc. 184. The Vopson factor  $\xi_{\text{FFGFT}}/\xi_{\text{UIFT}} \approx 414,684$  discussed in Doc. 013 appears, in the present picture, as a projection ratio between a layer-1 quantity ( $\xi_{\text{FFGFT}}$ , dimensionless) and a layer-2 quantity ( $\xi_{\text{UIFT}} = k_B T_{\text{CMB}} \ln 2 / (m_e c^2)$ , bound to the CMB temperature).

*Johann Pascher, Upper Austria, May 2026*

## References

- [1] J. Pascher, *Information as Topologically Frozen Kinetic Energy*, Dok. 255, Zenodo DOI: 10.5281/zenodo.20117635 (2026).
- [2] R. Landauer, *Irreversibility and heat generation in the computing process*, IBM Journal of Research and Development **5**(3), 183–191 (1961).
- [3] A. Bérut et al., *Experimental verification of Landauer's principle*, Nature **483**, 187–189 (2012).
- [4] M.M. Vopson, *The mass-energy-information equivalence principle*, AIP Advances **9**(9), 095206 (2019).